

SOLUTION PROPOSAL

Student Performance Analytics & Decision Support System

1. Proposed Solution Overview

Mega Central High School currently relies on manual spreadsheet-based reporting to monitor student performance. This approach leads to fragmented data, time-consuming report preparation, limited analytical capabilities, and delayed academic interventions.

The proposed solution is a centralized **Student Performance Analytics & Decision Support System** that integrates data collection, ETL processing, SQL-based data storage, business analytics, and interactive Power BI dashboards.

The solution automates the complete analytics workflow—from raw assessment data to actionable insights—enabling school management and teachers to make timely, data-driven academic decisions.

2. Solution Components

A. Data Layer

The Data Layer consists of raw educational datasets collected from multiple academic sources.

The system stores:

- Student Master Data
- Assessment/Test Data
- Question Bank
- Student Responses
- Test Schedule Information
- Subject & Chapter Information

Source Files

- students.csv
- assessments.csv
- question_bank.csv
- student_responses.csv
- tests.csv

B. ETL Processing Layer (Python)

The Python ETL pipeline is responsible for transforming raw data into analytics-ready datasets.

ETL Activities

- Data Extraction
- Data Cleaning
- Missing Value Handling
- Duplicate Removal
- Data Validation

- Score Calculation
- Percentage Calculation
- Grade Assignment
- Pass/Fail Classification
- Performance Band Classification
- Student Risk Classification
- Data Transformation
- Export of Clean Analytical Tables

C. SQL Database Layer (Microsoft SQL Server)

After ETL processing, the cleaned datasets are loaded into **Microsoft SQL Server**, which serves as the centralized data repository for analytics.

Database Responsibilities

- Store cleaned datasets
- Maintain relational data
- Support business queries
- Perform data validation
- Enable reporting
- Ensure data consistency

Database Objects

Tables

- students
- assessments
- question_bank
- student_responses
- tests

SQL Scripts

- Database Creation
- Business Queries
- Data Validation Queries

Business users can execute SQL queries to answer academic questions without opening Power BI.

Examples include:

- Subject-wise average performance
- Top-performing students
- High-risk student identification
- Chapter-wise accuracy
- Pass percentage by grade
- Attendance analysis
- Performance trends

D. Analytics Layer (Power BI)

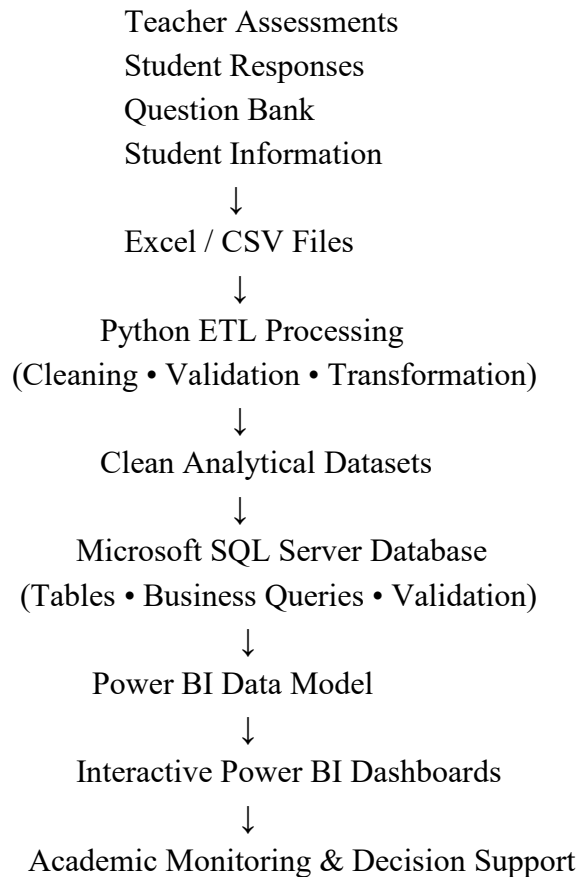
Power BI connects to the cleaned analytical data to create interactive dashboards.

The analytics layer provides:

- Executive Performance Dashboard
- Individual Student Dashboard
- Risk Analysis Dashboard
- Subject & Chapter Performance Dashboard

Power BI enables interactive filtering, drill-down analysis, KPI monitoring, and data visualization.

3. Solution Architecture



4. Expected Outcomes

Current State	Proposed Future State
Manual spreadsheet reporting	Automated interactive dashboards
Disconnected Excel files	Centralized SQL database
Time-consuming report generation	Real-time analytics
Limited academic insights	Data-driven decision making

Delayed identification of weak students	Early risk detection
Manual performance calculations	Automated KPI calculations
Difficult trend analysis	Interactive trend monitoring
Separate departmental reports	Unified analytics platform

5. Implementation Approach

Phase 1 – Business Analysis

Activities

- Stakeholder Identification
- Requirement Gathering
- Business Process Analysis
- BRD Preparation
- FRD Preparation
- User Stories
- Process Flow Design

Phase 2 – Data Preparation

Activities

- Collect raw datasets
- Perform Python ETL
- Clean and validate data
- Generate analytical datasets

Phase 3 – SQL Database Development

Activities

- Create SQL Database
- Import cleaned datasets
- Create database tables
- Perform data validation
- Execute business queries
- Verify data quality

Phase 4 – Dashboard Development

Activities

- Design Power BI data model
- Create DAX measures
- Build dashboards
- Configure interactions

- Validate visualizations

Phase 5 – Testing & Deployment

Activities

- Functional Testing
- Data Validation Testing
- User Acceptance Testing (UAT)
- Defect Resolution
- Final Dashboard Deployment
- User Documentation
- Project Closure

6. Technology Stack

Component	Technology
Programming Language	Python
ETL	Pandas, NumPy
Database	Microsoft SQL Server
Query Language	SQL
Reporting & Visualization	Microsoft Power BI
Documentation	Microsoft Word
Presentation	Microsoft PowerPoint
Version Control	GitHub

7. Key Benefits

The proposed solution provides the following benefits:

- Centralized academic data management
- Automated data cleaning and processing
- Reliable SQL-based data storage
- Improved data accuracy through validation
- Faster academic reporting
- Interactive performance dashboards
- Early identification of at-risk students
- Chapter-level learning gap analysis
- Better curriculum planning
- Data-driven academic decision making
- Scalable architecture for future expansion

8. Future Enhancements

The proposed solution can be extended with additional capabilities such as:

- Parent Notification System
- Teacher Performance Analytics
- Mobile Application
- Student Self-Service Portal
- Predictive Performance Analytics
- AI-Based Learning Recommendations
- Attendance Integration
- Learning Management System (LMS) Integration
- Real-Time Dashboard Refresh
- Automated Email Alerts for High-Risk Students